WEEK 4 – CYBERSECURITY INTERNSHIP

Advanced Networking, Analysis & Security Concepts

Mini Project Report

Submitted By: Name: Amaan Fayaz
Role: Cybersecurity Intern

Internship Domain: Cybersecurity

Week: Week 4

Project Title

Basic Network Security Lab using Kali Linux, Ubuntu, Nmap & Wireshark

Tools Used

- Kali Linux
- Ubuntu Linux
- UTM
- Nmap
- Wireshark
- Apache Web Server

Project Components

- Network Setup
- Service Configuration
- Nmap Network Scanning
- Wireshark Traffic Analysis
- Mini Project Workflow
- Advanced Nmap Scan

Task 1: Setup Basic Network (Lab)

Objective: To create a small isolated cybersecurity laboratory using two virtual machines, Kali Linux and Ubuntu Linux, and establish network connectivity between them for authorized security testing.

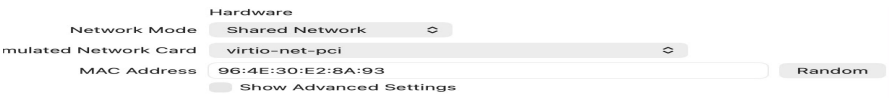
Lab Environment:

Machine	Role	IP address	Network
Kali Linux	Attacker	192.168.64.2	Shared
Ubuntu 24.04.5 LTS	Target	192.168.64.3	Shared

Network Configuration: Both Kali Linux and Ubuntu Linux were configured in UTM using Shared Network mode. The Kali Linux machine was used as the attacker system, while Ubuntu Linux was configured as the target system. Both machines received IP addresses from the same 192.168.64.0/24 network, allowing them to communicate with each other.

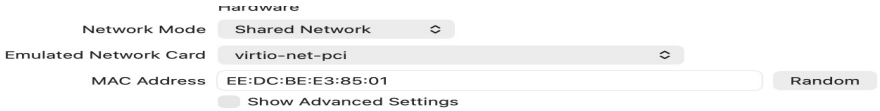
Connectivity Testing: Network connectivity was verified from Kali Linux using the ping command. Four ICMP packets were sent to the Ubuntu target. All four packets were successfully received, resulting in a 0% packet loss. This confirmed that the attacker and target machines could communicate successfully. Command: ping -c 4 192.168.64.3

Fig 1. Kali Network Settings



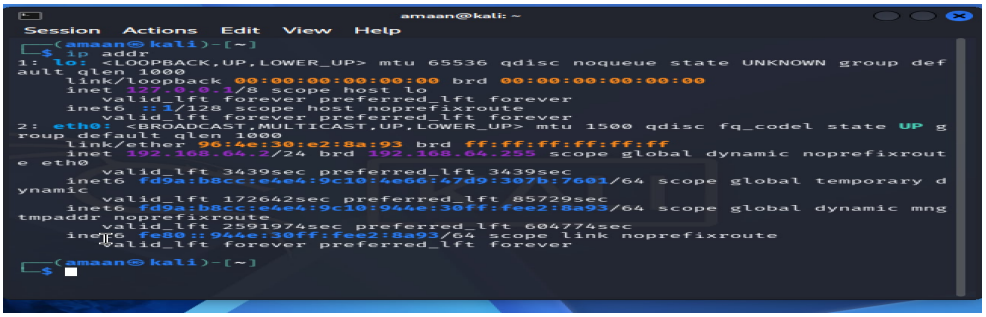
Kali Linux was configured as the attacker machine using Shared Network mode.

Fig 2. Ubuntu Network Configuration



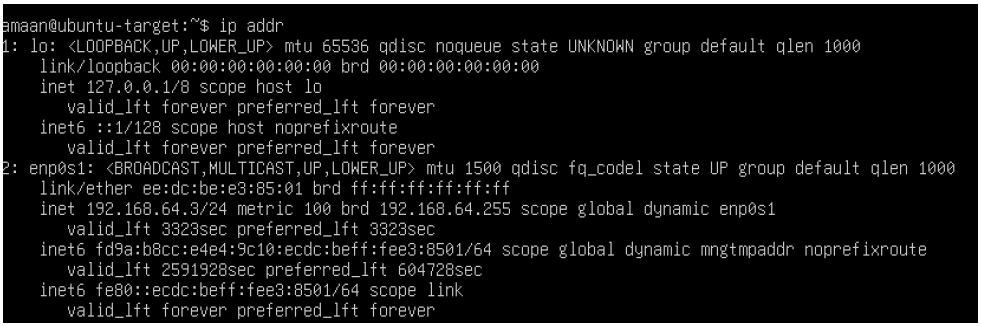
Ubuntu Linux was configured as the target machine using Shared Network mode.

Fig 3. IP address configuration of Kali Linux attacker machine



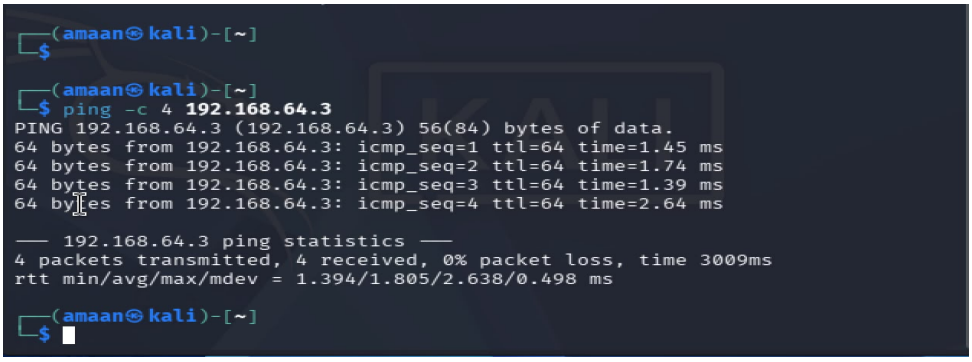
Kali Linux was assigned the IPv4 address 192.168.64.2/24 and configured as the attacker machine.

Fig 4. IP address configuration of Ubuntu Linux target machine



The Ubuntu target machine was assigned the IPv4 address 192.168.64.3/24. This address allows it to communicate with the Kali Linux attacker machine on the same virtual network.

Fig 5. Successful Connectivity Test



The ping test was successful with 0% packet loss, confirming communication between the Kali attacker and Ubuntu target.

Task 2: Install & Configure Services

For this task, Apache Web Server was installed and configured on the Ubuntu target machine. Apache was selected to provide a basic HTTP service for the security testing lab. The system packages were first updated using sudo apt update. Apache was then installed and started on the Ubuntu target machine.

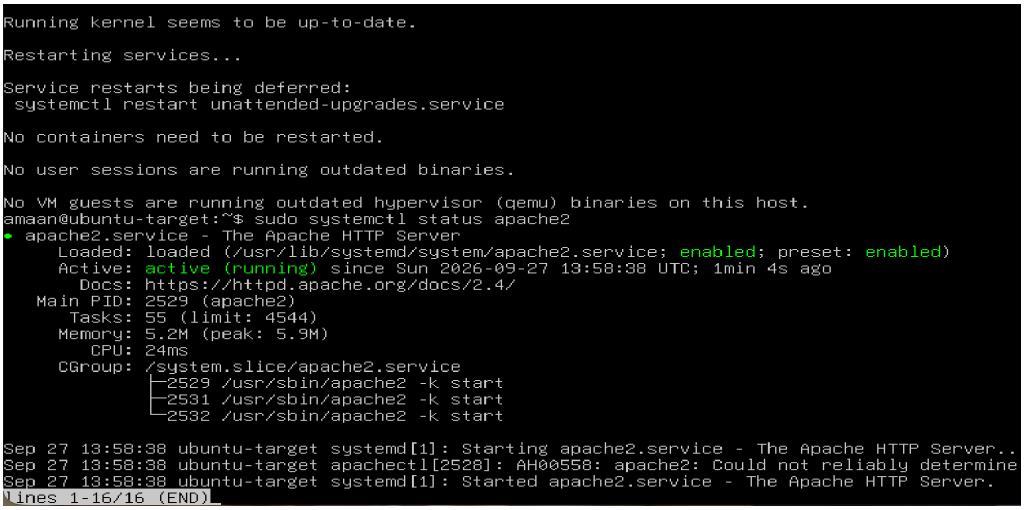
The Apache service was checked using the following command:

sudo systemctl status apache2

The output showed that the Apache service was active (running), confirming that the web server was successfully installed and running.

The target machine's IP address was: 192.168.64.3

Fig. 6. Apache Web Server Running on Ubuntu Target



Apache HTTP Server was successfully installed and is running on the Ubuntu target machine.

Task 3 = Network Scanning with Nmap

Objective

The objective of this task was to perform network scanning of the Ubuntu target machine using Nmap from Kali Linux. The scan was used to identify whether the target was reachable, discover open ports, and determine the services and versions running on the target.

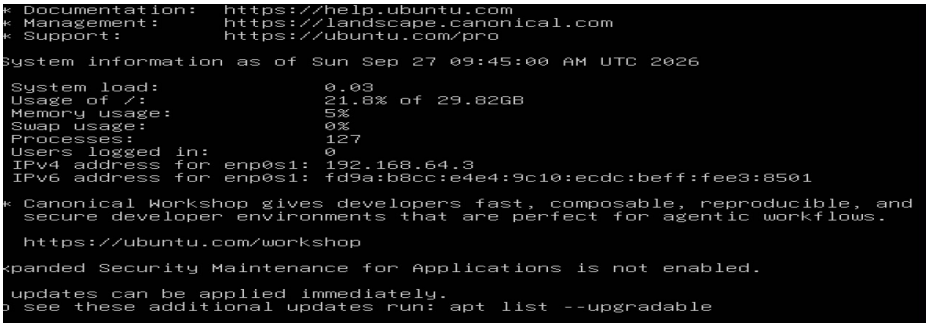
Target Information

The Ubuntu target machine had the following IPv4 address:

Target IP Address: 192.168.64.3

The target machine was running Ubuntu Linux and was configured as the target system for the cybersecurity lab.

Fig. 7. Ubuntu Target Machine Network Information



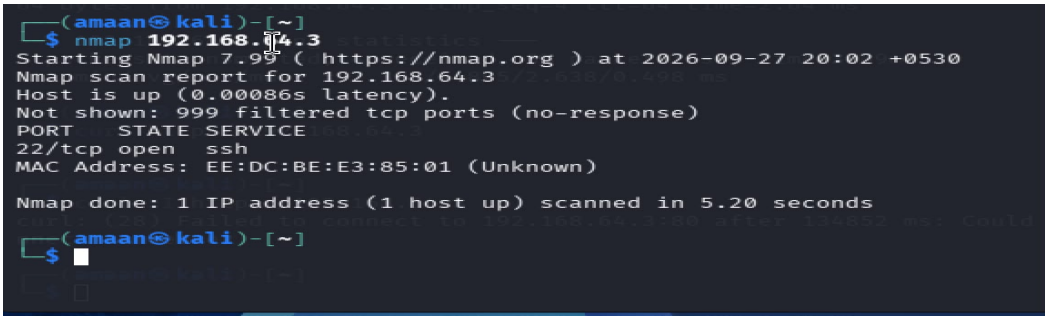
The Ubuntu target machine was identified with the IPv4 address 192.168.64.3 on the enp0s1 network interface. This IP address was used as the target for the Nmap scans performed from Kali Linux. The system information confirmed that the Ubuntu machine was active and available for the cybersecurity lab.

Basic Nmap Scan

A basic Nmap scan was performed from the Kali Linux machine to identify the open ports and services available on the Ubuntu target. The following command was used: nmap 192.168.64.3

The scan confirmed that the Ubuntu target was up and reachable. Port 22/tcp was found to be open and was identified as the SSH service. The remaining 999 TCP ports were reported as filtered.

Fig. 8. Basic Nmap Scan of Ubuntu Target



The basic Nmap scan confirmed that the Ubuntu target at 192.168.64.3 was up and reachable. Port 22/tcp was found open, with SSH running as the associated service. The remaining 999 TCP ports were reported as filtered.

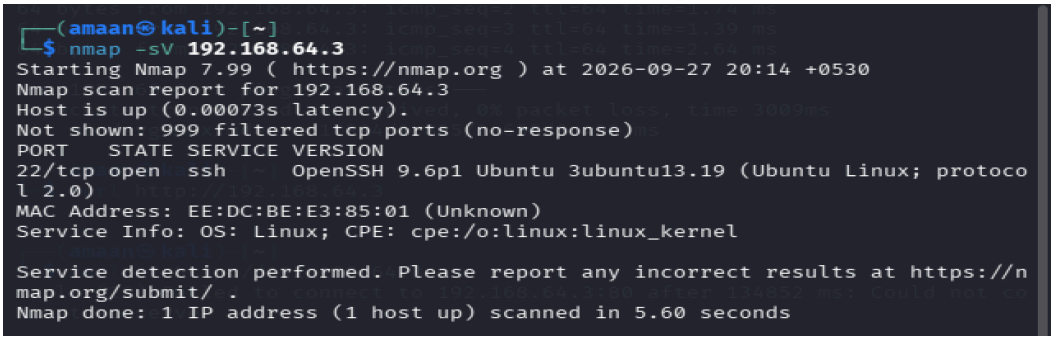
Basic Nmap Scan Result

The basic Nmap scan was performed from Kali Linux against the Ubuntu target at 192.168.64.3. The scan confirmed that the target machine was up and reachable.

The scan identified 22/tcp as an open port, with SSH running as the associated service. The remaining 999 TCP ports were reported as filtered, meaning Nmap did not receive a response from those ports.

The following command was used: nmap -sV 192.168.64.3

Fig. 9. Nmap Service and Version Detection Scan of Ubuntu Target



The Nmap service and version scan identified 22/tcp as an open port running the SSH service. The detected version was OpenSSH 9.6p1 Ubuntu 3ubuntu13.19, and the target system was identified as Linux. This provided useful information about the service running on the Ubuntu target.

Task 4 – Wireshark Traffic Capture.

Objective

The objective of this task was to use Wireshark to capture and analyze network traffic between the Kali Linux attacker machine and the Ubuntu target machine. ICMP, HTTP, and DNS traffic were captured and examined using Wireshark display filters.

Wireshark Setup

Wireshark was installed and opened on the Kali Linux attacker machine. The eth0 network interface was selected because it was the active interface connected to the virtual network. Packet capture was performed while generating different types of network traffic.

ICMP Traffic Capture

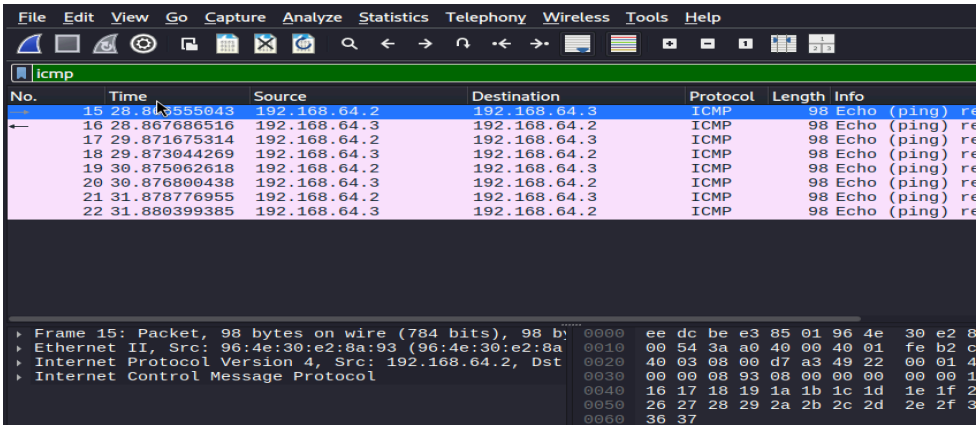
An ICMP ping test was performed from Kali Linux to the Ubuntu target using:

ping -c 4 192.168.64.3

The traffic was captured in Wireshark and filtered using:

Icmp

Fig. 11. ICMP Traffic Capture Between Kali Linux and Ubuntu Target



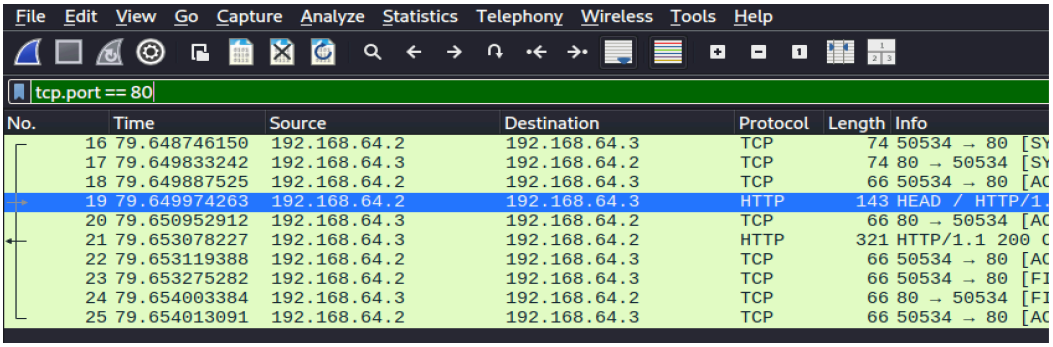
Wireshark successfully captured ICMP Echo Request and Echo Reply packets between Kali Linux (192.168.64.2) and the Ubuntu target (192.168.64.3). This confirmed that the communication between the two virtual machines was successfully captured and analyzed.

HTTP Traffic Capture

HTTP traffic was generated by accessing the Apache web server running on the Ubuntu target using: curl -I --connect-timeout 5 http://192.168.64.3:80

The captured traffic was filtered using: tcp.port == 80

Fig. 12. HTTP Traffic Capture of Apache Web Server



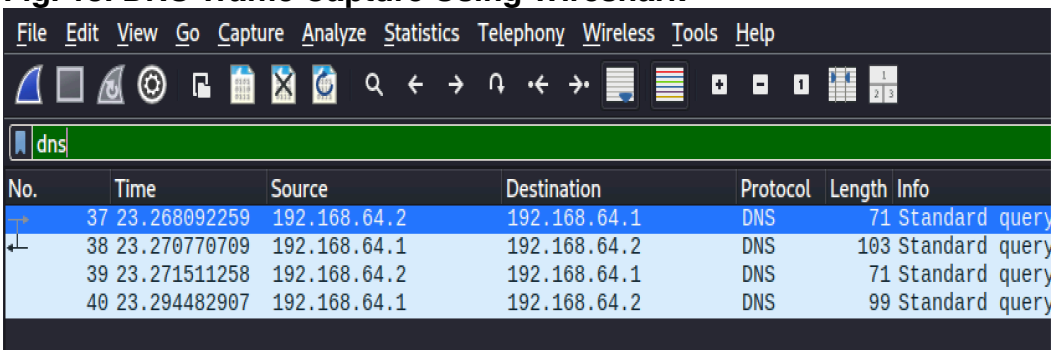
The capture showed communication between Kali Linux (192.168.64.2) and Ubuntu (192.168.64.3) over TCP port 80. The packets included an HTTP HEAD request and an HTTP/1.1 200 OK response, confirming successful communication with the Apache web server.

DNS Traffic Capture

A DNS lookup was performed from Kali Linux using: nslookup example.com

The captured packets were filtered using: dns

Fig. 13. DNS Traffic Capture Using Wireshark



Wireshark captured DNS query and response packets between the Kali Linux machine (192.168.64.2) and the DNS resolver (192.168.64.1). This demonstrated how DNS requests and responses can be observed using Wireshark.

Task 4 Conclusion: Wireshark was successfully used to capture and analyze ICMP, HTTP, and DNS traffic in the cybersecurity lab. The captures demonstrated how different network protocols can be identified and examined using appropriate Wireshark display filters.

Task 5 – Mini Project Output

Objective

The objective of this mini project was to demonstrate a complete basic network security testing workflow using Kali Linux as the attacker machine and Ubuntu Linux as the target machine. The project covered network setup, service configuration, Nmap scanning, and network traffic analysis using Wireshark.

Complete Workflow

The mini project was completed through a series of practical steps. Kali Linux was used as the attacker and testing machine, while Ubuntu Linux was configured as the target machine. Both machines were connected through the same UTM Shared Network.

Network Setup → Services Running → Nmap Scan → Traffic Capture → Analysis

Workflow Stage	Evidence
Network Setup	Fig 5. successful connectivity test
Services Running	Fig 6. Apache Web Server
Scan Results	Fig 9. Nmap Service and Version Detection
Traffic Capture	Fig 12. HTTP Traffic Capture

Network Setup

Kali Linux was assigned the IP address 192.168.64.2, while the Ubuntu target was assigned 192.168.64.3. Both virtual machines were connected through UTM Shared Network. Connectivity was verified successfully using ICMP ping with 0% packet loss.

Services Running

Apache Web Server was installed and configured on the Ubuntu target. The Apache service was confirmed to be active and accessible through TCP port 80. HTTP connectivity from Kali was verified using a curl request, which returned an HTTP/1.1 200 OK response.

Scan Results

Nmap was used from Kali Linux to scan the Ubuntu target. The scan identified TCP port 22 as open with SSH running. Service and version detection identified OpenSSH 9.6p1 Ubuntu 3ubuntu13.19. The Apache HTTP service on port 80 was also verified separately through HTTP testing.

Traffic Capture

Wireshark was used to capture and analyze network traffic generated during the practical activities. ICMP traffic was captured during the ping test, HTTP traffic was captured during communication with the Apache web server, and DNS traffic was captured during a DNS lookup.

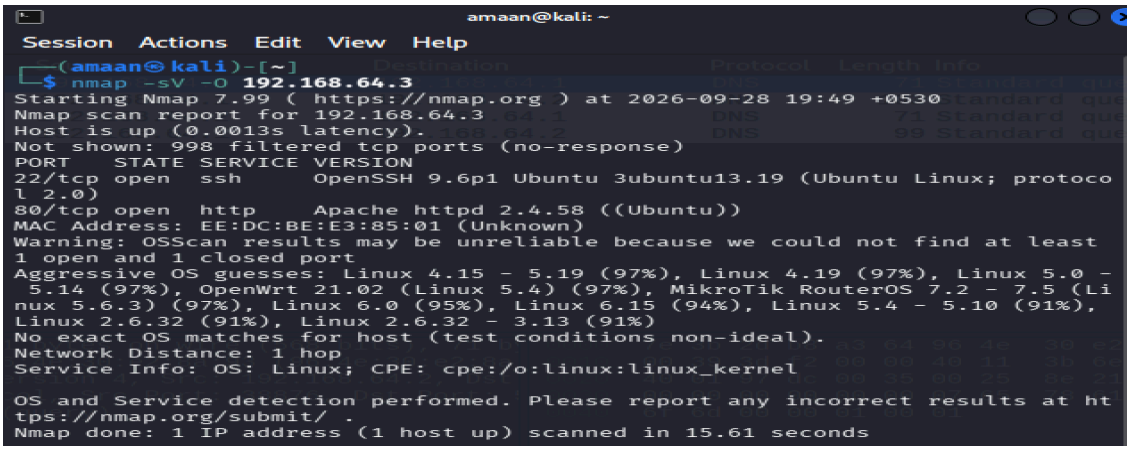
Mini Project Findings

The practical exercise demonstrated how a basic network security assessment can be performed in a controlled virtual environment. Network connectivity was established, services were identified, Nmap was used for service discovery, and Wireshark was used to observe network traffic. The results provided practical understanding of network scanning, service identification, and packet analysis.

Task 5 Conclusion

The mini project successfully demonstrated the complete workflow from network setup to traffic analysis. The combination of Kali Linux, Ubuntu Linux, Nmap, Apache Web Server, and Wireshark provided a practical environment for learning basic network security assessment techniques.

Fig. 14. Advanced Nmap Service and Operating System Detection



Advanced Nmap Scan

An advanced Nmap scan was performed against the Ubuntu target using service/version detection and operating-system detection. The command nmap -sV -O 192.168.64.3 was used to gather additional information about the services and operating system of the target machine.

The scan provided additional information about the services running on the Ubuntu target and attempted to identify the underlying operating system. This demonstrated the use of advanced Nmap options for network reconnaissance in the controlled laboratory environment.

Conclusion

This activity demonstrated how Nmap's advanced options can provide additional information about network services and the operating-system family of a target in a controlled cybersecurity laboratory.